

Alignment tests on pillar type drilling machine



* **Properly Installed** – In both horizontal (longitudinal and transverse directions) and vertical directions.

1. Flatness of clamping surface of base:

- * Straight edge on two gauge blocks
- * Feeler gauges gives error
- * Error should not exceed $0.1/1000$ mm clamping surface
- * Surface should be concave only

2. Flatness of clamping surface of table:

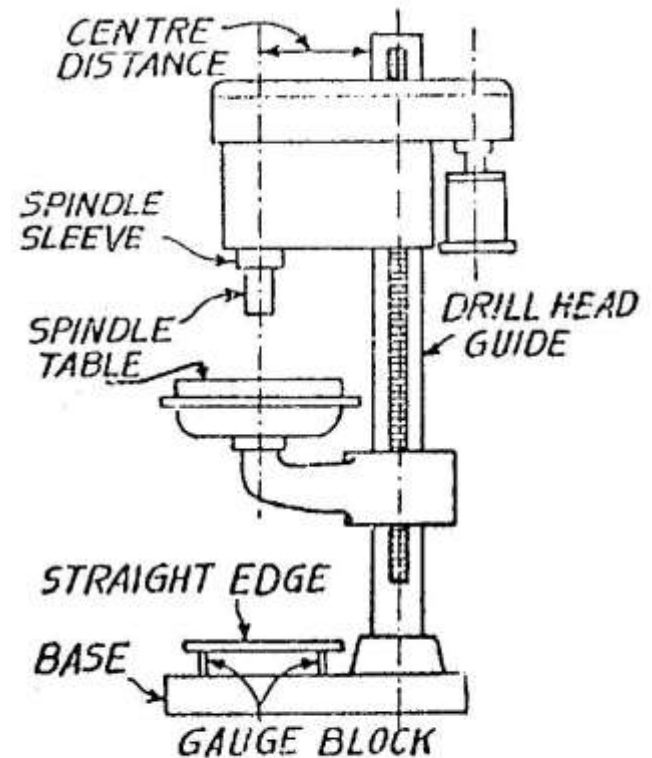


Fig. 16.17. Checking flatness of clamping surface of base.

3. Perpendicularity of drill head guide to the base plate :

- * Tested in both vertical plane(a) and in a plane at 90° to the above plane (b)
- * Frame level with graduations from 0.03 to 0.05 mm/m
- * Error should not exceed 0.25/1000 mm for (a) and 0.15/1000 mm for (b)

4. Perpendicularity of drill head guide with table :

5. Perpendicularity of spindle sleeve with base plate :

- * Tested in both the planes using frame level
- * Error should not exceed 0.25/1000 mm for plane (a) and 0.15/1000 mm for plane (b)

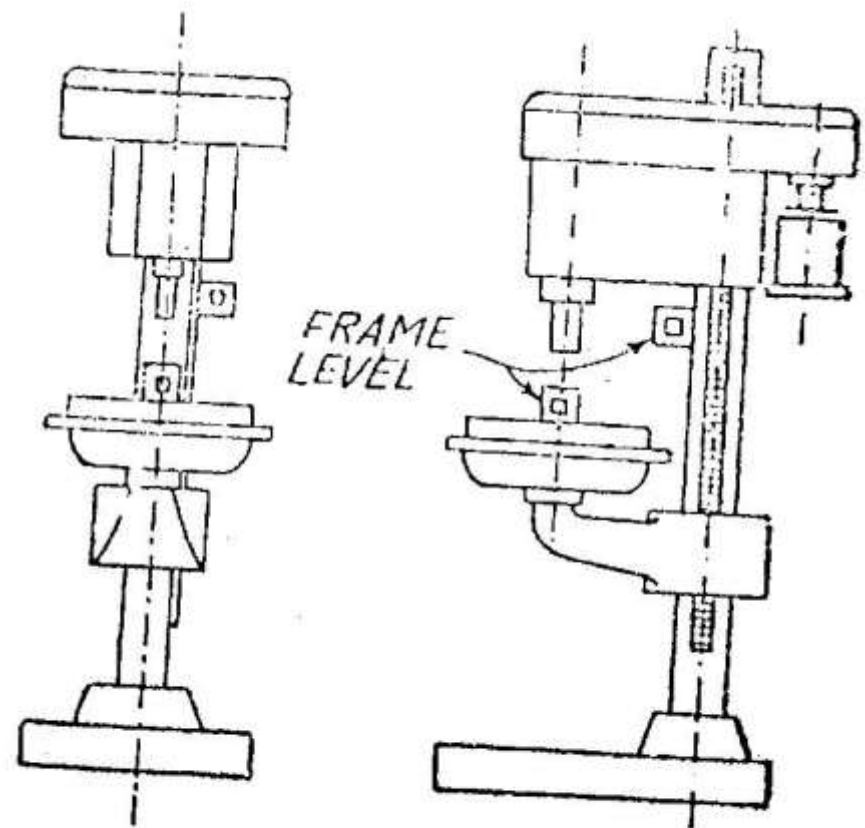


Fig. 16.18. Test for perpendicularity of drill head with table.

6. True running of spindle taper :

- * Test mandrel is placed in the tapered hole of spindle
- * Dial indicator
- * Spindle rotated slowly
- * Error should not exceed 0.03/100 mm for machines with taper upto Morse No. 2 and 0.04/300 mm for machines with taper larger than Morse No. 2

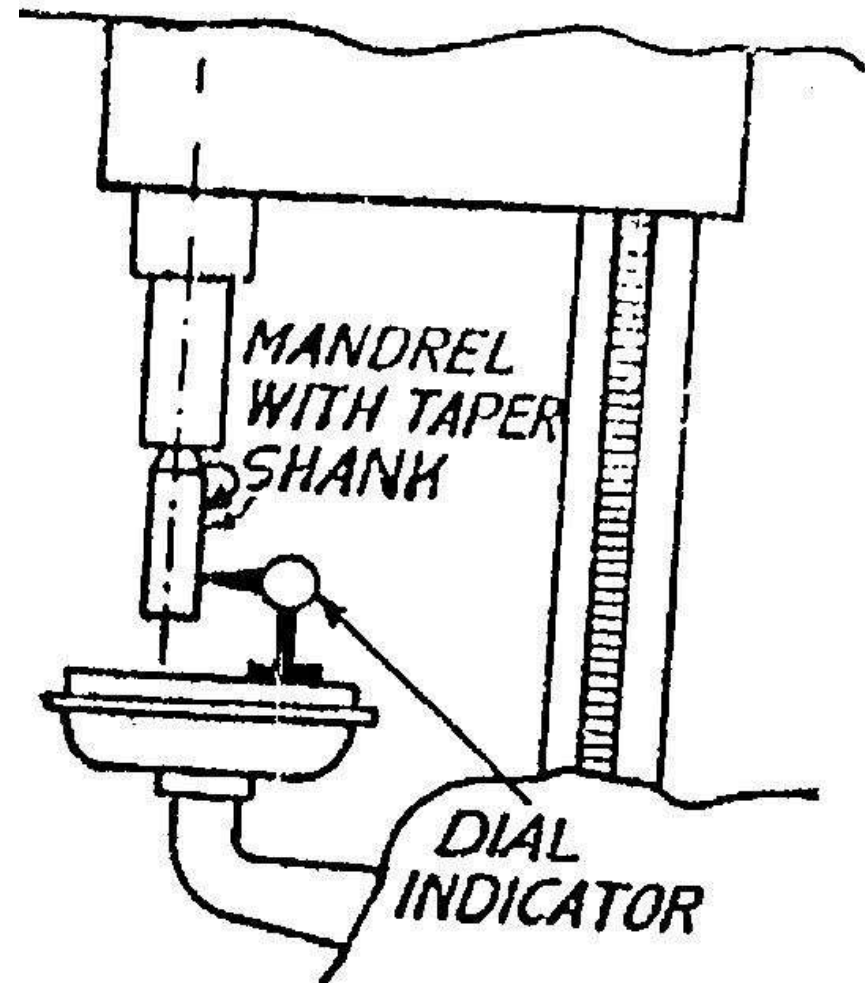


Fig. 16.19

7. Parallelism of the spindle axis with its vertical movement :

- * Test is performed into two planes A & B
- * Test mandrel and dial indicator
- * Spindle is adjusted in the middle position of its travel
- * Permissible errors are : For machines with taper upto Morse No. 2 0.03/100 mm for plane A and 0.03/100 mm for plane B.
- * For machines with taper larger than Morse No. 2 0.05/300 mm for plane A and 0.05/300 mm for plane B.

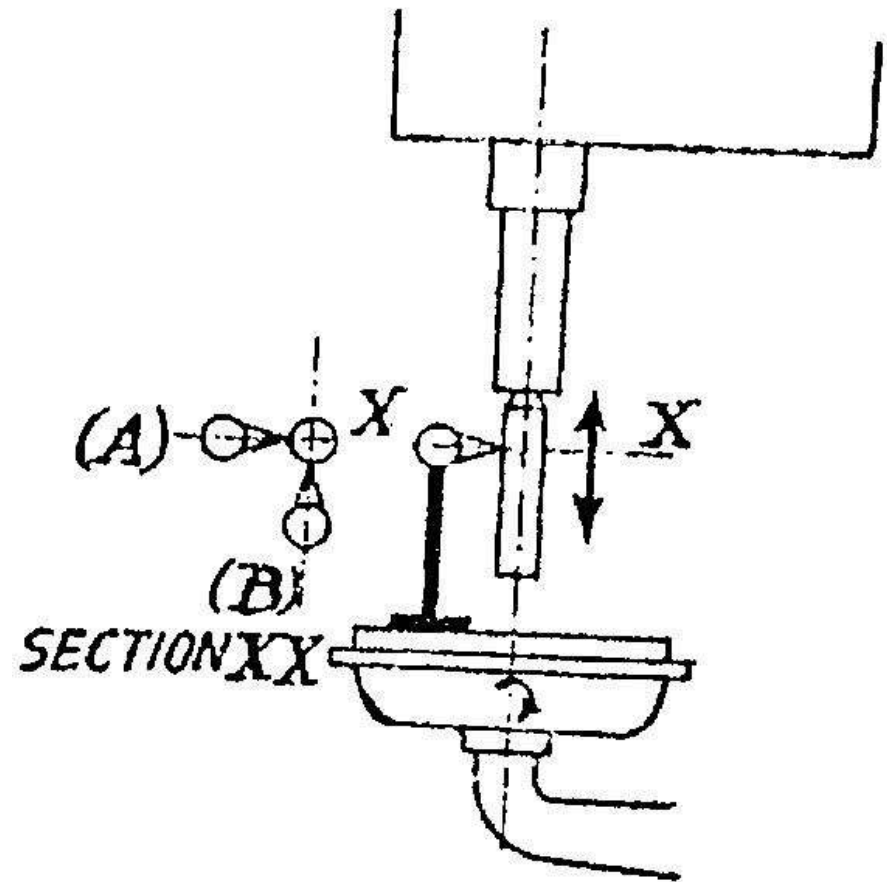


Fig. 16.20

8. Squareness of clamping surface of table to its axis :

- * Dial indicator
- * Table is slowly rotated
- * Error should not exceed 0.05/300 mm diameter

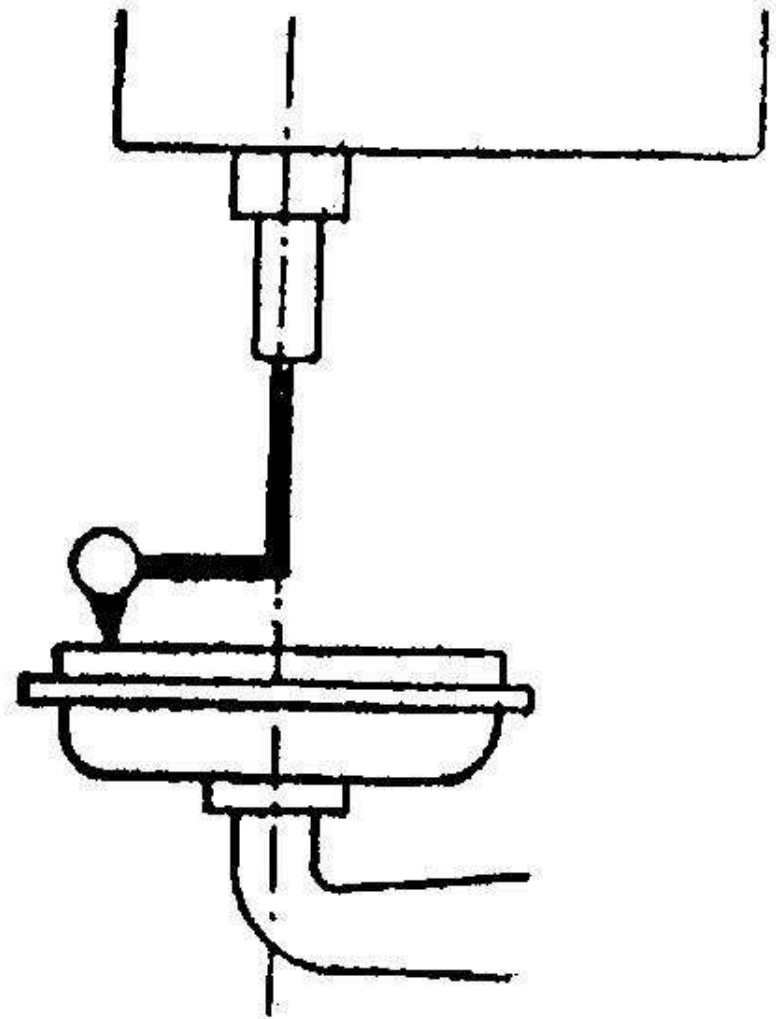


Fig. 16.21

9. Squareness of spindle axis with table :

- * Straight edge placed in position AA' and BB'
- * Work table in middle position of its vertical travel
- * Dial indicator
- * $A - 180^\circ - A'$
- * The permissible errors are 0.08/300 mm for AA' and 0.05/300 mm for BB'.

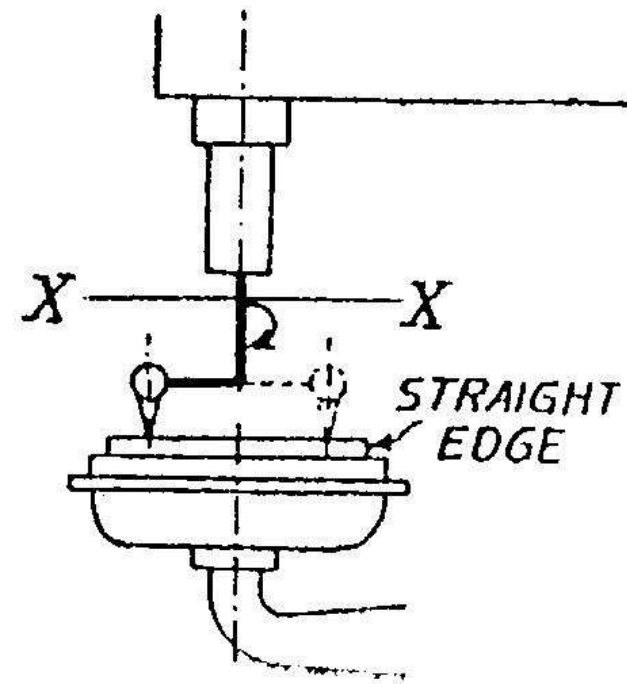
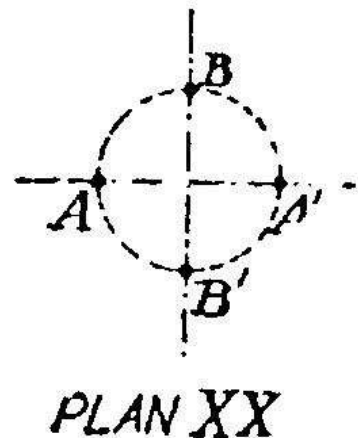


Fig. 16.22

10. Total deflection :

- * Drill head and table are arranged in their middle position.
- * Dial indicator
- * Dynamometer

<i>Drilling Diameter in Open Hearth Steel</i>	<i>kg</i>
6	100
10	200
16	350
20	550
25	750
32	950
40	1200

	<i>for centre distance upto and including</i>
0.4 mm	200 mm
0.6 mm	200-300 mm
0.8 mm	300-400 mm
1.0 mm	400 and above mm

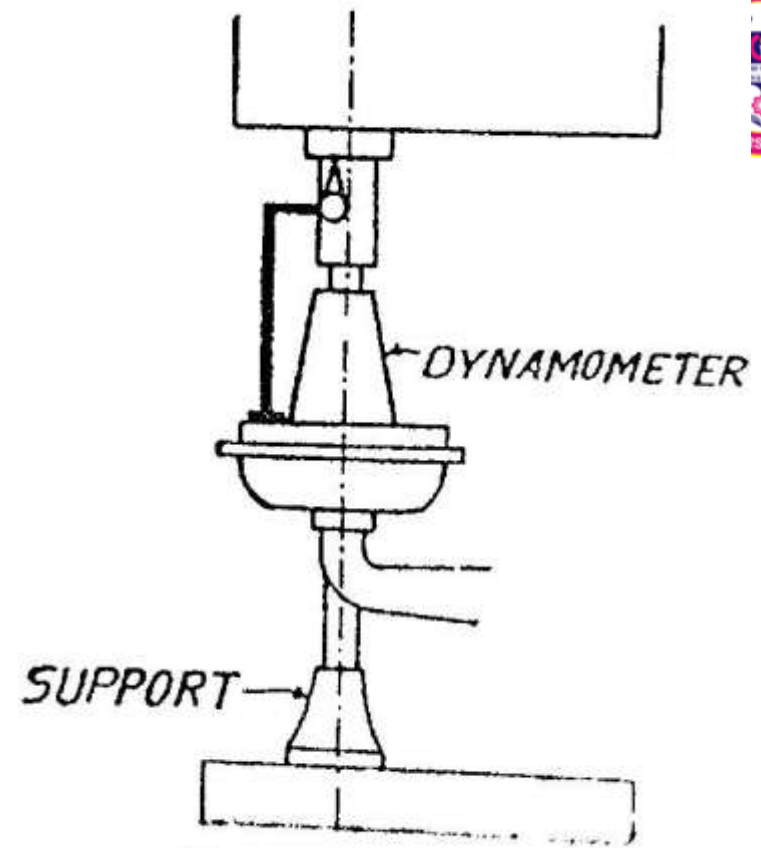


Fig. 16.23